

Coupled effect of freeze–thaw and chloride immersion cycles on bond–slip behavior of steel and FRP rebars in concrete

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ABSTRACT

This experimental study investigates the bond–slip behavior of steel and FRP rebars in concrete under the coupled effect of freeze–thaw (FT) and chloride immersion (CI) cycles. While both FT and CI can individually diminish load-carrying capacity by reducing the bond strength, their combined effect on rebars with different material properties and surface conditions remains insufficiently explored. To address this, a thorough experimental investigation was conducted on the bond–slip behavior of steel, in-production helically wrapped glass fiber reinforced polymer (IHW-GFRP), and post-production helically wrapped carbon fiber reinforced polymer (PHW-CFRP) rebars embedded in concrete. A total of 36 cubic concrete specimens reinforced with these rebars were subjected to up to 300 FT cycles and 5 CI cycles, followed by pull-out testing to evaluate the combined effect of these cycles on bond–slip behavior. Additionally, 18 cylindrical concrete specimens underwent the same conditions and were tested under compression to assess changes in compressive strength. The experimental results revealed that the combined FT and CI cycles reduced the peak bond stress by 14.63 %, 16.46 %, and 43.51 % for steel, IHW-GFRP, and PHW-CFRP rebars, respectively. In addition, the CEB-FIP analytical model for FRP rebars was modified to enhance the accuracy in estimating bond–slip curves of specimens subjected to the combined FT and CI cycles. This modification provided an improvement compared to the estimated curves using the conventional CEB-FIP model.

1. Introduction

Rebars, both steel and fiber reinforced polymer (FRP), are crucial for reinforcing concrete structures, improving their tensile strength and durability. Steel rebars, made from carbon steel, offer high elastic moduli and tensile strengths, addressing concrete's inherent weakness in tension [1,2]. FRP rebars, which gained significant popularity recently, provide a high strength-to-weight ratio, corrosion resistance, and low thermal conductivity [3–7]. The robust bond between these rebars and concrete is essential for ensuring structural integrity and performance. This bond, facilitated by chemical adhesion, friction, and mechanical interlocking forces, is influenced by various factors including type and surface condition of rebars, their diameter, concrete compressive strength, embedment length, concrete cover, and environmental conditions [8–14]. Among these factors, environmental conditions including freeze–thaw (FT) and chloride immersion (CI) cycles significantly affect bond–slip behavior and the bond strength of rebars embedded in concrete. FT cycles cause the spread of microcracks due to water expansion and contraction within concrete pores, weakening the

bond between concrete and rebars. Li et al. [15] investigated the bond–slip behavior of steel rebars in different concrete types and found that after 200 FT cycles, peak bond stress decreased by 54 %, 30 %, and 26 % for natural aggregate, half natural and half recycled aggregate, and fully recycled aggregate concretes, respectively. Shakiba et al. [16] studied the bond–slip behavior of steel and glass fiber reinforced polymer (GFRP) bars in various concrete types under FT and wet–dry cycles. They observed peak bond stress reductions of 5 %, 2 %, and 4 % for steel bars embedded in normal, self-compacting, and fiber reinforced concretes, respectively, after 200 FT cycles. Similarly, GFRP bars experienced reductions of 2 %, 1 %, and 4 % in peak bond stress when embedded in normal, self-compacting, and fiber reinforced concretes, respectively, following the same number of FT cycles. Ma et al. [17] evaluated the effect of FT cycles with sustained loading on bond–slip behavior between steel rebars and various types of concrete. It was concluded that the peak bond stress of specimens under sustained loading of 30 % and 50 % of drawing force decreased by 22.8 % and 48.5 %, respectively, after 50 FT cycles. Alves et al. [18] tested the individual and combined effects of FT cycles along with sustained and

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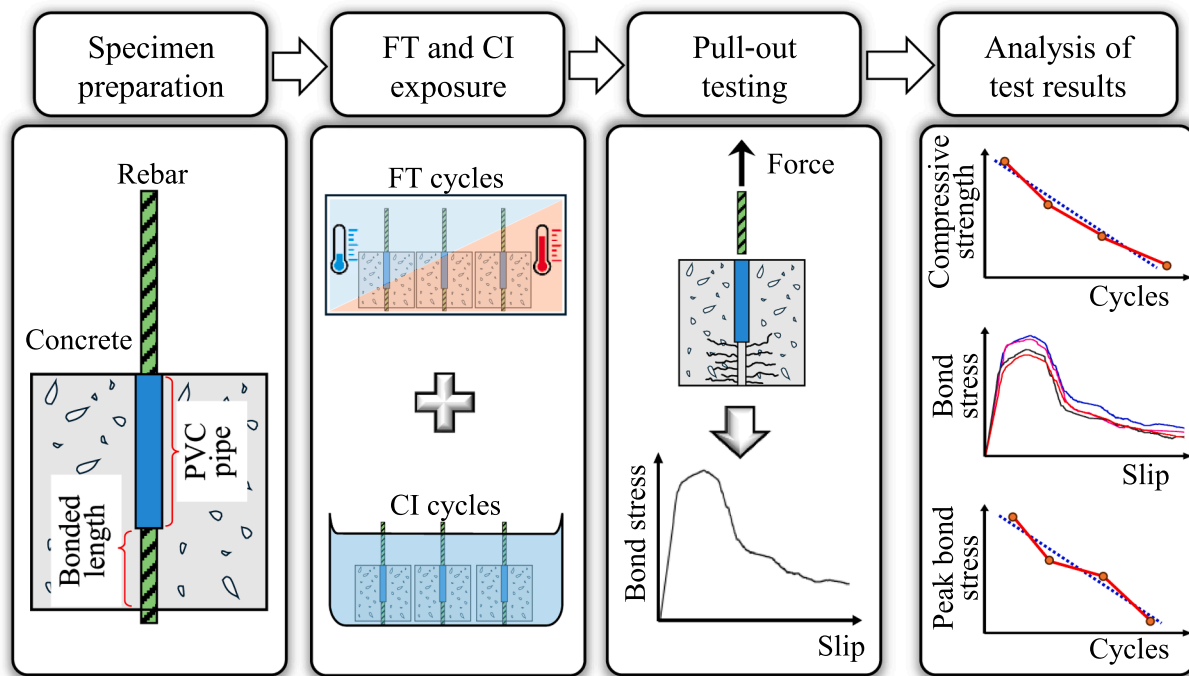


Fig. 1. Schematic illustration of the proposed experimental program for determining the bond-slip behavior under the combined influence of FT and CI cycles.

fatigue loading conditions on bond-slip behavior between GFRP bars and concrete. According to their results, FT cycles enhanced the bond strength by approximately 40 %.

Steel rebars in concrete inevitably undergo corrosion when exposed to chloride infiltration. Chloride ions, commonly from sources such as deicing salts and seawater, permeate concrete, reaching steel rebars and initiating corrosion. This process leads to the formation and expansion of corrosion products, creating microcracks in concrete, potentially weakening the bond between rebars and concrete [19]. Conversely, in concrete reinforced with FRP rebars, which are naturally resistant to corrosion, chloride exposure weakens the bond by degrading the polymer matrix of FRP and reacting with certain cement hydration products to form Friedel's salt [20,21]. Studies conducted by Shang et al. [22] showed that under chloride attack alone, bond strength of steel rebars increased slightly but decreased under combined sustained load and chloride attack. Chang et al. [23] examined GFRP bars in seawater sea-sand concrete under both seawater immersion and sustained load conditions. They found that in the case of 12 mm diameter GFRP rebars, the peak bond stress initially increased by 4.8 %, but it gradually declined by 7.4 % over time. In contrast, 16 mm diameter GFRP rebars consistently improved the peak bond stress, reaching a value 15.7 % higher than the initial measurement. Wenjie et al. [24] evaluated the impact of chloride corrosion and sustained stress on the bond strength between steel-FRP composite bars and concrete. They observed an initial bond strength increase for steel-basalt FRP rebars submerged in sodium chloride solution for up to 135 days, followed by a decline. Conversely, bond strength for steel-glass FRP rebars consistently decreased throughout the testing period. Bazli et al. [25] explored the influence of seawater, alkaline, and acidic environments on the bond behavior of GFRP bars in different concrete types. They found minimal loss in peak bond stress, measuring less than 7 %, for self-compacting and high-strength concretes submerged in seawater, while lightweight and normal concretes experienced higher reductions in peak bond stress, reaching up to 14 %.

Several studies have extensively examined the separate impacts of FT and CI cycles, yet their combined influence remains poorly understood. Su et al. [26] aimed to bridge this gap by studying how salt-frost cycles affect the bond behavior between steel rebars and recycled coarse

aggregate concrete. They observed a significant 55 % decrease in peak bond stress after 125 cycles. Ren et al. [27] studied bond-slip behavior between recycled aggregate concrete and steel bars under rapid FT cycles in both fresh and saline water, finding maximum peak bond stress losses of 33.5 % and 44.7 % after 75 cycles, respectively. Belarbi et al. [28] examined the combined effect of FT cycles, high temperature cycles, and deicing salt solution on bond strength, focusing on GFRP and CFRP rebars in various concrete types. They observed a larger decrease in peak bond stress for GFRP rebars, amounting to a 56 % loss, compared to CFRP rebars, which experienced a 23 % reduction. Yan et al. [29] explored the bond durability of GFRP bars in concrete under harsh environmental conditions, including FT cycles and alkaline-saline solution with 3 % sodium chloride concentration, observing a 14 % reduction in peak bond stress for concrete specimens with small covers and 8 % for those with large covers.

While these studies highlight the effect of combined environmental conditions, they reveal significant discrepancies in findings due to differences in simulated environmental conditions, materials, and testing methods. To address this, a comparative experimental study is conducted to evaluate the bond-slip behavior of three types of rebars under the influence of combined FT and CI cycles. Concrete specimens embedded with steel, in-production helically wrapped GFRP (IHW-GFRP), and post-production helically wrapped CFRP (PHW-CFRP) rebars are subjected to up to 300 FT cycles and 5 CI cycles, followed by direct pull-out testing to assess bond-slip behavior of each rebar. Additionally, this study analytically estimates the bond-slip curves of specimens exposed to different levels of FT and CI cycles by determining the peak and residual bond stress variations caused by these simulated environmental conditions. A modification to the conventional CEB-FIP model for FRP rebars is also proposed to improve the accuracy of estimating the bond-slip relationship and address discrepancies between experimental and analytical results.

2. Experimental program

Among factors that affect the bond strength between rebars and concrete, FT and CI cycles pose significant issues, such as rebar corrosion, microcrack propagation, and deterioration of certain hydration

Table 1

Chemical composition and mechanical properties of PC 42.5 cement.

Chemical composition (%)										Compressive strength (MPa)	
CaO	SiO ₂	Al ₂ O ₃	Fe ₂ O ₃	SO ₃	MgO	K ₂ O	Na ₂ O	Cl	Loss in ignition	3 days	28 days
65.08	19.70	4.97	3.58	2.65	0.91	0.77	0.17	0.011	2.15	34	52

Table 2

Concrete mix compositions.

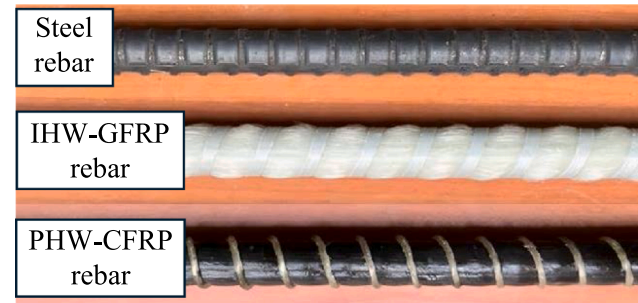
Materials	Weight (kg/m ³)
Cement	503
Water	202
Coarse aggregate	905
Fine aggregate	715
Air entrainer	12.5

products of concrete. Although these two environmental conditions individually affect the bond-slip relationship, their coupled effect potentially poses a much more significant influence on reinforced concrete elements. Fig. 1 illustrates the proposed experimental program to determine the bond-slip behavior of various rebars with different surface conditions under the coupled effect of FT and CI cycles. In this proposed method, following the casting of concrete, all specimens undergo curing process. Subsequently, the cured specimens are exposed to combined FT and CI cycles. Once these cycles are completed, the specimens are subjected to pull-out testing. In the final step of the proposed approach, the combined impact of FT and CI cycles on compressive strength of concrete, the bond-slip relationship between concrete and rebars, and the bond strength between concrete and rebars are analyzed based on the obtained results from compressive and pull-out tests.

2.1. Materials

2.1.1. Concrete

In accordance with recommendations from the American Concrete Institute ACI 318-19 [30], the target compressive strength was established at 40 MPa, surpassing the minimum design strength of 35 MPa necessary for concrete exposed to environments with high chloride content. PC42.5 cement, similar to ASTM Type III Portland cement with chemical components and mechanical properties outlined in Table 1, was selected for its superior strength and durability properties. The concrete design mixture, detailed in Table 2, comprised 202 kg/m³ water, 503 kg/m³ cement, 905 kg/m³ coarse aggregate, and 715 kg/m³ sand, with a water-to-cement ratio of 0.4. To enhance workability and attain the desired air content, a 2.5 % air entrainer was added to the mix. According to ASTM C192/C192M-14 standard [31], the mixing process started by adding a portion of water and half of the aggregates to the mixer. Next, the cement was added, followed by the rest of the aggregates, and finally, the remaining water and admixtures. Once mixed, the concrete was carefully poured into specialized molds, consolidated using rodding. The batch was then stored at room temperature for 24 h to prevent moisture loss during the initial curing phase. To mitigate the negative effects of winter weather on the curing process of concrete elements and enhance early age compressive strength development, the specimens were removed from the molds and placed in a water tank maintained at 65 °C for one week to accelerate early age curing. Following this phase, the water temperature was gradually reduced to 23 °C within a day and then held constant until the testing day, thereby completing the curing process. After curing, 3 cylindrical concrete specimens measuring 100 mm in diameter and 200 mm in height underwent a compressive strength test. The average compressive strength of this specific mixture was measured as 42.6 MPa with a standard deviation of 1.24 MPa.

**Fig. 2.** Rebars employed in this study.**Table 3**

Mechanical properties of rebars.

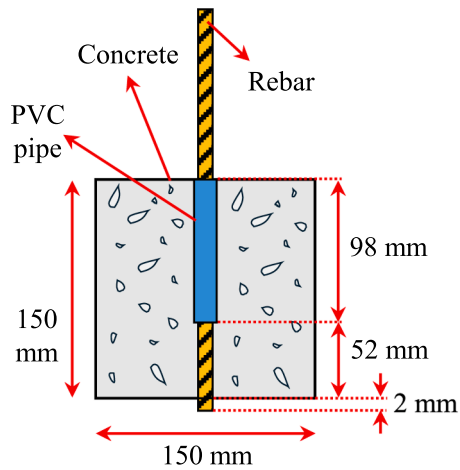
Rebar type	Tensile strength (MPa)	Modulus of elasticity (GPa)	Nominal bar diameter (mm)	Cross-sectional area (mm ²)
Steel	440	200	12.7	126.7
IHW-GFRP	1000	60	12.7	132
PHW-CFRP	2000	145	12.7	113

2.1.2. Rebars

As demonstrated in Fig. 2, the study employed steel, IHW-GFRP, and PHW-CFRP rebars to assess bond-slip behavior under the combined influence of FT and CI cycles. These rebars offered different surface conditions depending on their manufacturing procedures. The steel rebars were manufactured as typical ribbed steel rebars. The IHW-GFRP rebars, produced by PSS Engineering, were manufactured by tightly wrapping a fiber bundle around the GFRP rebar during its production, forming rigid ridges along its surface. The PHW-CFRP rebars, manufactured by the Korea Institute of Ceramic Engineering and Technology, were initially produced as CFRP rebars with smooth surfaces. Subsequently, an additional fiber bundle was helically wrapped and adhered to their surfaces. Therefore, steel and IHW-GFRP rebars were categorized as rebars with deformed surfaces, while PHW-CFRP rebars were classified as rebars with semi-deformed surfaces. To standardize the analysis and eliminate the influence of bar diameter on bond-slip behavior, all rebars were set to a diameter of 12.7 mm. Further details regarding the mechanical properties of these rebars can be found in Table 3.

2.2. Specimen design

The concrete specimens were prepared in accordance with Annex G of the Canadian standard CAN/CSA-S806-12 [32], which prescribes a reliable and standardized test method for determining the bond strength of FRP rods embedded in concrete by pull-out testing. As shown in Fig. 3, the specimens were cubes with each side measuring 150 mm. Rebars were centrally aligned through holes in the mold to maintain uniform positioning within the concrete. The embedment length of rebars plays a crucial role in achieving their potential bond strength. When the embedment length is too short, premature pull-out failure may occur, resulting in suboptimal bond strength. Conversely, an excessively long embedment length can lead to localized failure of the rebar or concrete



(a)



(b)

Fig. 3. Specimens designed for pull-out testing: (a) schematic view of specimen's vertical section and (b) cured specimens.

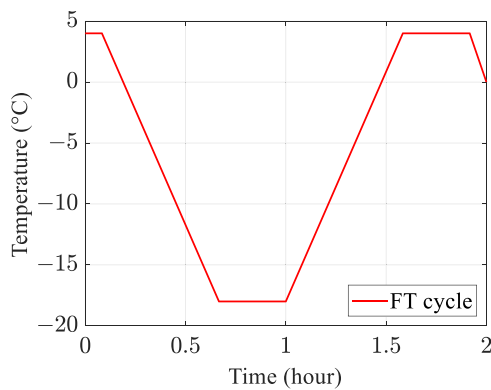
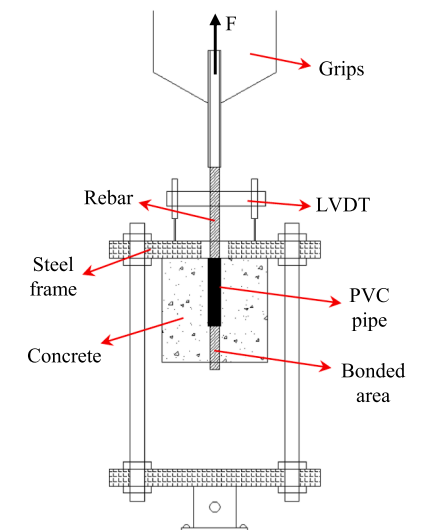


Fig. 4. Temperature profile of 1 FT cycle.

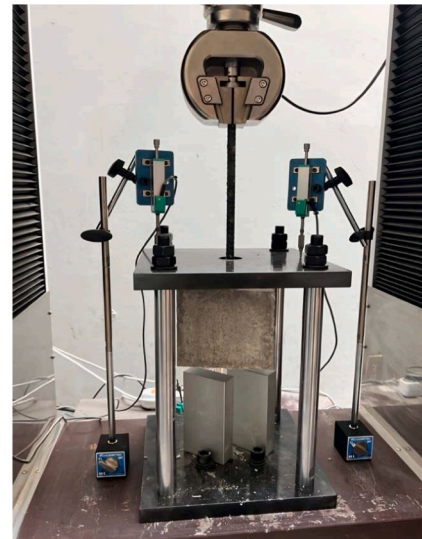
due to uneven stress distribution and stress concentration near the loaded end, which similarly reduces the achievable bond strength [33]. To mitigate these risks, the embedment length was set at 52 mm, exceeding four times the nominal rebar diameter as specified by the CAN/CSA-S806-12 standard. The remaining 98 mm of the rebar remained unbonded using a PVC pipe. To examine the bond-slip behavior under the coupled influence of FT and CI cycles, 36 specimens were prepared.

2.3. Simulated environmental conditions

In this study, the environmental conditions of FT and CI were simulated by following specific protocols outlined in ASTM C666/C666M-03 [34] and the Nordic standard NT build 443 [35], respectively. According to ASTM C666/C666M-03, the concrete specimens embedded with rebars underwent 300 FT cycles, with temperatures at each cycle ranging from -18°C to 4°C and lasting for 2 h. As indicated in Fig. 4, the temperature was initially set at 4°C and maintained for 5 min. Subsequently, a freezing phase commenced, gradually lowering



(a)



(b)

Fig. 5. Pull-out test setup: (a) schematic view and (b) experimental apparatus.

Table 4
Specimens prepared for pull-out testing.

Specimens	Rebar type	No. of combined cycles
S-0-1	Steel	Control specimens (0 FT and 0 CI cycles)
S-0-2		
G-0-1	IHW-GFRP	
G-0-2		
C-0-1	PHW-CFRP	
C-0-2		
S-1-1	Steel	1 (60 FT and 1 CI cycles)
S-1-2		
G-1-1	IHW-GFRP	
G-1-2		
C-1-1	PHW-CFRP	
C-1-2		
S-2-1	Steel	2 (120 FT and 2 CI cycles)
S-2-2		
G-2-1	IHW-GFRP	
G-2-2		
C-2-1	PHW-CFRP	
C-2-2		
S-3-1	Steel	3 (180 FT and 3 CI cycles)
S-3-2		
G-3-1	IHW-GFRP	
G-3-2		
C-3-1	PHW-CFRP	
C-3-2		
S-4-1	Steel	4 (240 FT and 4 CI cycles)
S-4-2		
G-4-1	IHW-GFRP	
G-4-2		
C-4-1	PHW-CFRP	
C-4-2		
S-5-1	Steel	5 (300 FT and 5 CI cycles)
S-5-2		
G-5-1	IHW-GFRP	
G-5-2		
C-5-1	PHW-CFRP	
C-5-2		

the temperature to -18°C over 35 min, where it remained for an additional 20 min. Following this, a thawing phase began, with the temperature rising back to 4°C over 35 min, sustained for an additional 20 min. Finally, the temperature was lowered to 0°C over 5 min, completing one cycle.

The simulation of chloride exposure was conducted using a sodium chloride solution with a concentration of 165 g NaCl per liter, following the guidelines outlined in the Nordic standard NT build 443. Each CI cycle began with the complete submersion of the specimens in this solution at room temperature to ensure that all surfaces were exposed to the chloride ions. To accelerate the penetration of chloride ions, the specimens remained immersed for 2 days. Subsequent to this immersion phase, the specimens underwent a controlled drying process by being placed in a heating chamber set at 60°C for an additional day. This completed one cycle.

The combined effect of FT and CI cycles on the specimens was simulated by introducing CI cycles between FT cycles. Specifically, after every 60 FT cycles, a CI cycle was applied to the specimens. This resulted in five combined FT and CI cycles, with cycle 1 comprising 60 FT and 1 CI cycles, cycle 2 comprising 120 FT and 2 CI cycles, cycle 3 comprising 180 FT and 3 CI cycles, cycle 4 comprising 240 FT and 4 CI cycles, and cycle 5 comprising 300 FT and 5 CI cycles.

2.4. Pull-out testing

The pull-out tests followed the CAN/CSA-S806-12 Annex G Canadian standard, which offers precise guidelines for evaluating the bond strength of FRP rebars. To conduct the tests, a universal testing machine (UTM) with movable grips and a maximum capacity of 100 kN was utilized. As illustrated in Fig. 5, cubic concrete specimens each containing an embedded rebar were securely held in position by a metallic

frame. This frame's base was firmly affixed to a rigid base plate to ensure stability. To prevent any eccentricity in the applied pull-out force and to account for potential slippage of the metallic frame, two linear variable differential transformer (LVDT) sensors were positioned on the frame's top surface. The rebars were pulled by the UTM grip at a constant speed of 1.27 mm/min. Following the recommendations outlined in CAN/CSA-S806-12, the load and displacement data were captured at a sampling rate of 2 Hz, and by assuming a uniform stress distribution along the bonded length of rebars, the bond stress was calculated using Eq. (1) provided below.

$$\tau = \frac{F}{\pi dl} \quad (1)$$

where τ is the bond stress, F is the tensile force, d is the bar diameter, and l is the bonded length. As outlined previously, a total of 36 specimens were prepared for bond strength analysis through the pull-out testing. These specimens were categorized into 3 groups, with each group comprising 12 specimens representing a specific type of rebar. Among these, 10 specimens in each group underwent combined FT and CI cycles, while 2 specimens served as control specimens and were not exposed to these cycles. Table 4 provides details regarding the specimens designated for pull-out testing. In this table, specimens embedded with steel, IHW-GFRP, and PHW-CFRP were labeled as S, G, and C respectively. Furthermore, the number of combined FT and CI cycles and the specimen count were indicated following the letter representing the rebar type in sequential order. For instance, C-3-2 denotes the second specimen embedded with PHW-CFRP and subjected to 3 combined FT and CI cycles.

3. Coupled effect of FT and CI

The results of both compressive and pull-out tests are presented in Table 5, where the peak bond stress and its corresponding slippage are denoted as $\tau_{\max}$ and $s_{\max}$, respectively. These results were further analyzed in subsequent sections, including an examination of concrete's compressive strength, an assessment of the bond-slip behavior of rebars utilizing the experimentally obtained bond-slip curves, and an evaluation of bond strength in rebars subjected to combined FT and CI cycles.

3.1. Compressive strength of concrete

To establish the correlation between the compressive strength of concrete and the combined effect of FT and CI cycles, 3 cylindrical specimens, matching the dimensions used for determining the compressive strength of the designed mixture, underwent a compressive test at the end of each combined FT and CI cycle. Fig. 6 illustrates the relationship between the compressive strength of concrete specimens and the number of combined FT and CI cycles. The figure demonstrates a consistent reduction in compressive strength as the number of cycles increases. Specifically, after 1, 2, 3, 4, and 5 combined FT and CI cycles, the reductions in compressive strength were 4.62 %, 7.91 %, 9.79 %, 12.04 %, and 15.09 % respectively. The primary causes of this reduction are the propagation of cracks and degradation of the concrete matrix due to exposure to combined FT and CI cycles.

3.2. Bond-slip relationship

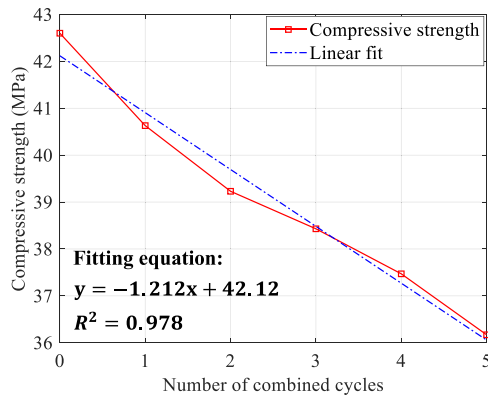
3.2.1. Steel rebars

Fig. 7 demonstrates the bond-slip curves of concrete specimens embedded with steel rebars. These curves were obtained by averaging the experimentally measured curves from each combined cycle and applying a low-pass filter to slightly smooth the results. The graph illustrates three distinct stages in the bond-slip behavior of control specimens and those exposed to up to 3 combined FT and CI cycles. These stages are identified as the ascending, plateau, and descending

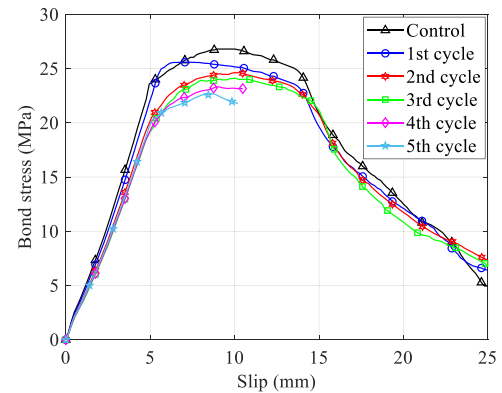
Table 5

Pull-out and compressive test results.

Specimens		$\tau_{\max}$ (MPa)	$s_{\max}$ (mm)	Failure mode	Concrete compression strength (MPa)	
					Tests	Average
Control specimens	S-0-1	28.19	10.89	Pullout	44.30	42.60
	S-0-2	25.41	9.03	Pullout	41.40	
	G-0-1	23.91	5.82	Pullout	42.10	
	G-0-2	22.87	5.34	Pullout		
	C-0-1	7.69	5.44	Pullout		
1 combined cycle	C-0-2	8.03	6.14	Pullout		40.63
	S-1-1	24.89	6.02	Pullout	40.50	
	S-1-2	26.27	7.72	Pullout	40.10	
	G-1-1	21.48	5.01	Pullout	41.30	
	G-1-2	23.86	6.09	Pullout		
2 combined cycles	C-1-1	7.18	6.07	Pullout		39.23
	C-1-2	7.00	5.77	Pullout		
	S-2-1	25.77	10.73	Pullout	39.40	
	S-2-2	23.45	9.19	Pullout	39.90	
	G-2-1	20.11	4.75	Pullout	38.40	
3 combined cycles	G-2-2	21.19	5.27	Pullout		38.43
	C-2-1	6.11	6.00	Pullout		
	C-2-2	6.71	6.52	Pullout		
	S-3-1	23.19	9.49	Pullout	38.60	
	S-3-2	25.03	10.63	Pullout	39.10	
4 combined cycles	G-3-1	20.93	5.89	Pullout	37.60	37.47
	G-3-2	19.38	4.48	Pullout		
	C-3-1	6.35	6.46	Pullout		
	C-3-2	5.59	5.90	Pullout		
	S-4-1	22.58	8.28	Split	37.70	
5 combined cycles	S-4-2	24.06	9.56	Split	38.20	36.17
	G-4-1	20.64	5.71	Pullout	36.50	
	G-4-2	18.94	4.91	Pullout		
	C-4-1	5.16	6.72	Pullout		
	C-4-2	4.45	5.96	Pullout		
5 combined cycles	S-5-1	24.27	8.65	Split	36.80	36.17
	S-5-2	21.49	7.53	Split	34.80	
	G-5-1	18.77	5.13	Split	36.90	
	G-5-2	20.31	5.95	Split		
	C-5-1	4.07	5.91	Pullout		
	C-5-2	4.81	6.25	Pullout		

**Fig. 6.** Relationship between the compressive strength of concrete specimens and the number of combined FT and CI cycles.

stages. In the ascending stage, both bond stress and slippage exhibited a nearly linear increase. Throughout this stage, the primary resistance to slippage is attributed to a combination of chemical adhesion, frictional resistance, and mechanical interlocking. During the plateau stage, bond stress slightly increased until reaching the peak bond stress, then slightly declined due to microcracking of the concrete near the rebar ribs. In the subsequent descending stage, there was a rapid decrease in bond stress due to localized concrete fracturing and shearing-off occurring between the rebar ribs. In contrast, specimens exposed to 4 and 5 combined FT and CI cycles exhibited only ascending and plateau stages, as the concrete underwent splitting due to extensive propagation of cracks

**Fig. 7.** Bond-slip curves of specimens embedded with steel rebars subjected to combined FT and CI cycles.

perpendicular to the longitudinal axis of rebars. This difference in behavior of specimens subjected to the different levels of FT and CI exposures is attributed to control specimens and those exposed to up to 3 cycles experiencing pull-out failure, as the concrete matrix was moderately affected by FT and CI cycles. Conversely, specimens exposed to 4 and 5 cycles experienced split failure due to significant cracking and deterioration of the concrete matrix from prolonged FT and CI exposure, as shown in Fig. 8.

3.2.2. IHW-GFRP rebars

The bond-slip curves of specimens embedded with IHW-GFRP rebars



Fig. 8. Split failure observed in concrete specimens embedded with steel rebars.

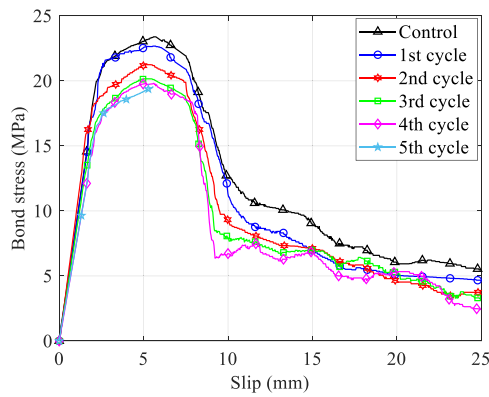


Fig. 9. Bond-slip curves of specimens embedded with IHW-GFRP rebars subjected to combined FT and CI cycles.

are depicted in Fig. 9. Similar to the previous section, these curves were derived by averaging and smoothing the experimentally measured data from each combined cycle. The figure illustrates that the bond-slip curves of specimens that underwent up to 4 combined FT and CI cycles exhibited three distinct stages: ascending, descending, and residual stages. During the ascending stage, bond stress increased nonlinearly with slip until reaching the peak bond stress. This stage relies on chemical adhesion, frictional resistance, and mechanical interlocking to resist slippage. Once the peak bond stress was reached, the descending stage started, characterized by a rapid decline in mechanical

interlocking due to localized concrete fracturing and shearing-off in the vicinity of rebar ridges, leading to a sharp decrease in bond stress. Upon entering the residual stage, significant slippage occurred due to fracture and sliding of concrete fragments between surface ridges of the rebar, accompanied by a gradual reduction in bond stress attributed to decreased frictional resistance. In contrast, specimens subjected to 5 combined FT and CI cycles exhibited only an ascending stage in their bond-slip curves since extensive propagation of cracks led to concrete splitting. This discrepancy is primarily attributed to varying failure modes resulting from different levels of deterioration induced by FT and CI cycles. Specifically, specimens exposed to up to 4 cycles experienced pull-out failure, while those exposed to 5 cycles experienced split failure, as illustrated in Fig. 10.

3.2.3. PHW-CFRP rebars

Fig. 11 illustrates the bond-slip curves of specimens embedded with PHW-CFRP rebars. Consistent with previous sections, these curves were derived by averaging and smoothing the experimental data from each combined cycle. All bond-slip curves of PHW-CFRP rebars exhibited three distinct stages: ascending, descending, and residual stages. Similar to IHW-GFRP rebars, during the ascending stage, the bond stress and slippage showed a nonlinear relationship until reaching the peak bond stress. At this stage, the resistance to slippage is a result of chemical adhesion, friction, and mechanical interlocking. In the descending stage following the peak bond stress, the first helically wrapped fiber bundle tore off, resulting in a significant reduction in mechanical interlocking, which ultimately caused a substantial decrease in bond stress. The residual stage showed minor ascending-descending patterns attributed to



Fig. 10. Split failure observed in concrete specimens embedded with IHW-GFRP rebars.

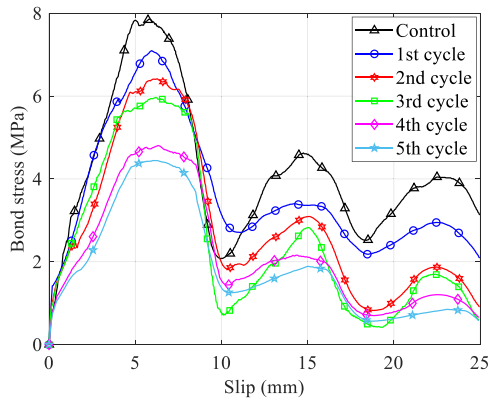


Fig. 11. Bond-slip curves of specimens embedded with PHW-CFRP rebars subjected to combined FT and CI cycles.

the engagement and dislodgement of remaining fiber bundles. These patterns resulted in peak bond stresses with smaller amplitudes compared to the primary peak bond stress observed during the ascending stage, and therefore, can be categorized as residual stress [36]. Notably, all specimens embedded with PHW-CFRP rebars failed through pull-out failure mode, as exemplified in Fig. 12. This indicates that even in the presence of severe concrete matrix deterioration, the helically wrapped fiber bundles failed before concrete cracking and fracturing occurred.

3.3. Bond strength

Upon comparing peak bond stress of control specimens presented in Table 5, PHW-CFRP rebars exhibited significantly lower peak bond stress compared to steel and IHW-GFRP rebars. This disparity can be attributed to the deformed surfaces of steel and IHW-GFRP rebars, which provide substantially higher mechanical interlocking due to their more rigid ribs and ridges. In contrast, the semi-deformed surface of PHW-CFRP rebars led to the dislodgement of the helically wrapped and adhered fiber bundle at relatively low stress levels, consequently reducing their mechanical interlocking capability. Fig. 13 illustrates the correlation between the combined FT and CI cycles and the peak bond stress of steel, IHW-GFRP, and PHW-CFRP rebars. According to this figure, the bond strength between rebars and concrete specimens subjected to combined FT and CI cycles decreased for all rebar types with an increase in the number of cycles. Compared to control specimens, the peak bond stress of steel rebars decreased by 4.55 %, 8.17 %, 10.04 %, 12.99 %, and 14.63 % after undergoing 1, 2, 3, 4, and 5 combined FT

and CI cycles, respectively. Similarly, IHW-GFRP rebars showed peak bond stress reductions of 3.08 %, 11.71 %, 13.83 %, 15.39 %, and 16.46 % after the same cycles, and PHW-CFRP rebars experienced decreases of 9.8 %, 18.45 %, 24.05 %, 38.87 %, and 43.51 % after the same cycles. According to these results, the combined effect of FT and CI cycles led to varying degrees of bond strength loss among different types of rebars. Steel and IHW-GFRP rebars showed the least peak bond stress losses after five cycles, with reductions of 14.63 % and 16.46 %, respectively, while IHW-GFRP rebars exhibited the highest peak bond stress loss after five cycles, experiencing a reduction of 43.51 %. The decrease in bond stress was consistent across all types of rebars and followed a uniform trend relative to the number of cycles. The relatively low peak bond stress losses of steel and IHW-GFRP rebars, and their split failure mode when subjected to a high number of combined FT and CI cycles, suggest that the deformed surfaces of these rebars were only marginally affected by the cycles. As a result, the mechanical interlocking forces were only slightly affected, leading to the split failure mode in the concrete matrix deteriorated by these cycles. Conversely, the significant bond strength loss of PHW-CFRP rebars at high numbers of combined FT and CI cycles, as well as the observation of the pull-out failure mode for specimens embedded with this type of rebar at the end of all cycles, indicate that the helically wrapped and adhered fiber bundle on the rebar surface dislodged at significantly lower stress levels. This early dislodgement, primarily due to the deterioration of the adhesive agent caused by the combined FT and CI cycles, resulted in pull-out failure within the deteriorated concrete matrix.

4. Analytical model

4.1. CEB-FIP model

In this study, the analytical estimation of bond-slip relationship was conducted using the CEB-FIP analytical model introduced by the fib Model Code [37]. This model employs separate analytical approaches to estimate the bond-slip relationship for steel and FRP rebars. For steel rebars, the CEB-FIP analytical model characterizes the bond-slip behavior through ascending, plateau, descending, and residual stages, as demonstrated in Fig. 14(a). This analytical model computes the bond-slip relationship based on the relative slippage (s), as shown in Eqs. (2a) to (2d) below.

$$\text{ascending stage : } \tau = \tau_{\max} \left(\frac{s}{s_1} \right)^{\alpha} \text{ for } 0 \leq s \leq s_1 \quad (2a)$$

$$\text{plateau stage : } \tau = \tau_{\max} \text{ for } s_1 < s \leq s_2 \quad (2b)$$

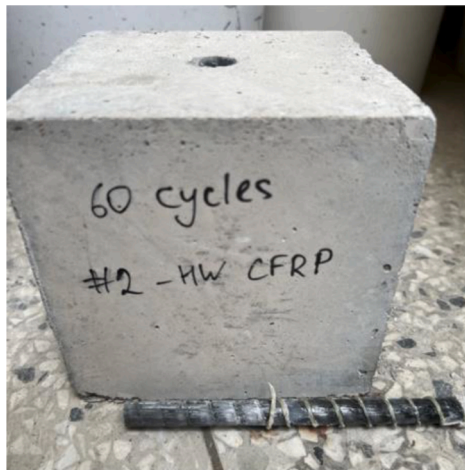


Fig. 12. Pull-out failure observed in concrete specimens embedded with PHW-CFRP rebars.

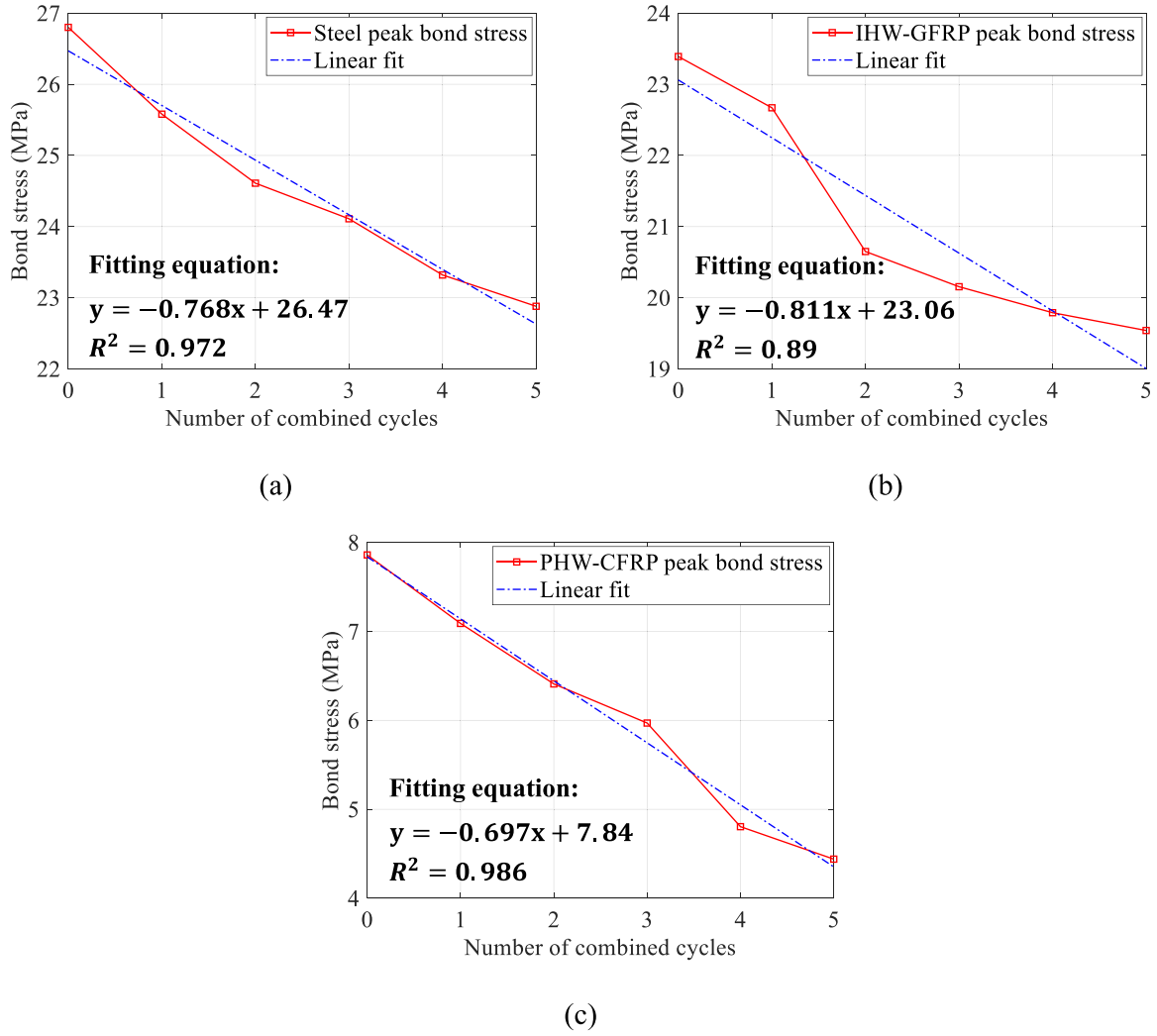


Fig. 13. Effect of combined FT and CI cycles on peak bond stress between concrete and (a) steel rebars, (b) IHW-GFRP rebars, and (c) PHW-CFRP rebars.

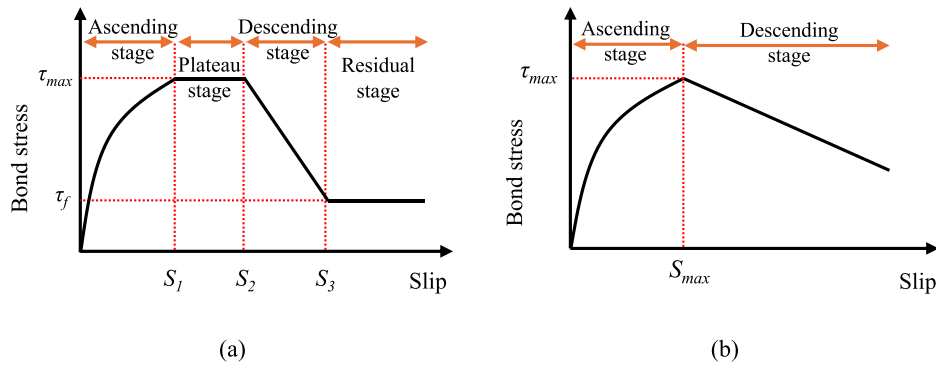


Fig. 14. CEB-FIP analytical model for: (a) steel rebars and (b) FRP rebars.

$$\text{descending stage : } \tau = \tau_{\max} - (\tau_{\max} - \tau_f) \left(\frac{s - s_2}{s_3 - s_2} \right) \text{ for } s_2 < s \leq s_3 \quad (2c)$$

$$\text{residual stage : } \tau = \tau_f \text{ for } s_3 < s \quad (2d)$$

where s_1 and s_2 are the slippages corresponding to the first and last points of the plateau stage, respectively. τ_f and s_3 are the residual bond stress and its corresponding slippage, respectively. α , which ranges from 0 to 1, is a parameter that determines the curvature of the initial

ascending stage in bond-slip relationship.

In the case of FRP rebars, the CEB-FIP analytical model adopts the modified BPE model proposed by Cosenza et al. [38,39]. As shown in Fig. 14(b), this model determines the bond-slip relationship by dividing it into ascending and descending stages. The calculation of this relationship involves the utilization of relative slippage, as illustrated in Eqs. (3a) and (3b) below.

Table 6

Experimentally measured curve fitting parameters for various types of rebars.

No. of combined cycles	Rebar type	α	p
Control specimens	Steel	0.9278	—
	IHW-GFRP	0.4048	−0.8765
	PHW-CFRP	0.6436	−0.1173
1 combined cycle	Steel	0.9525	—
	IHW-GFRP	0.3964	−0.8776
	PHW-CFRP	0.5849	−0.1685
2 combined cycles	Steel	0.9417	—
	IHW-GFRP	0.3851	−0.8122
	PHW-CFRP	0.6256	−0.2147
3 combined cycles	Steel	0.9310	—
	IHW-GFRP	0.4230	−0.7400
	PHW-CFRP	0.5081	−0.1910
4 combined cycles	Steel	0.9523	—
	IHW-GFRP	0.4433	−0.7153
	PHW-CFRP	0.5783	−0.1773
5 combined cycles	Steel	0.9423	—
	IHW-GFRP	0.4146	—
	PHW-CFRP	0.6381	−0.1682

$$\text{ascending stage : } \tau = \tau_{\max} \left(\frac{s}{s_{\max}} \right)^{\alpha} \text{ for } 0 \leq s \leq s_{\max} \quad (3a)$$

$$\text{descending stage : } \tau = \tau_{\max} - \tau_{\max} p \left(\frac{s - s_{\max}}{s_{\max}} \right) \text{ for } s_{\max} < s \quad (3b)$$

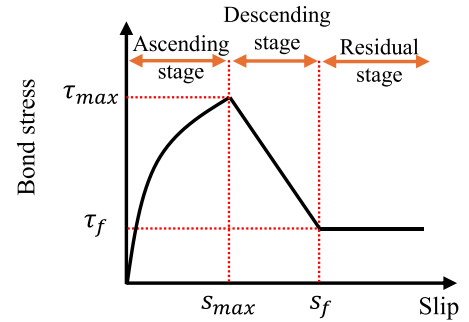
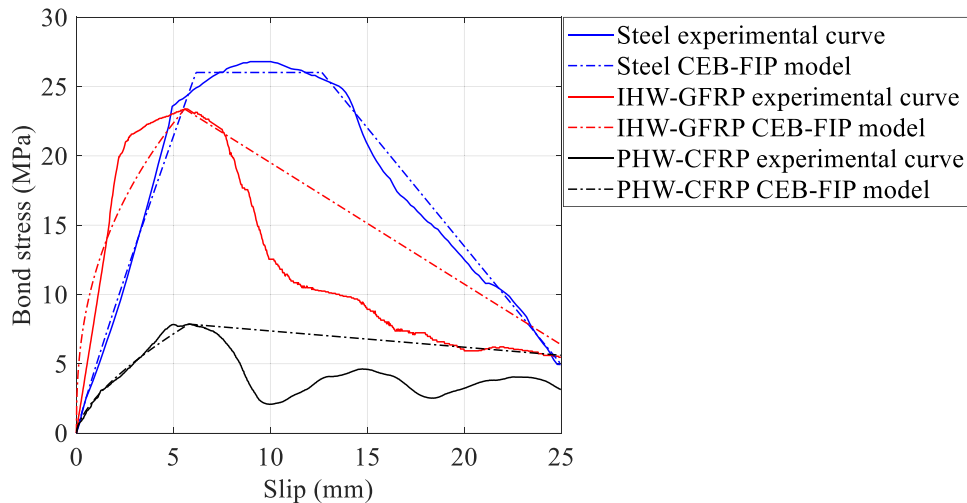
where p denotes the slope corresponding to the descending stage of the bond–slip curve. It is important to note that for parameters related to the analytical model of steel rebars, the fib Model Code recommends predetermined values for general design scenarios. In contrast, for parameters associated with the analytical model of FRP rebars, the model code recommends determining these parameters through regression analysis of experimental data. To ensure consistency in the determination of parameters for both steel and FRP rebar analytical models and to address potential mismatches between the CEB-FIP model's predetermined parameters for steel rebars and the experimental bond–slip curves, which may result from variations in design conditions and material characteristics, regression analysis of experimental data was employed. The experimentally measured values of α and p for all types of rebars are presented in Table 6. The table demonstrates slight and inconsistent variations in both parameters as the number of combined FT and CI cycles increases.

Fig. 15 illustrates the bond–slip curves of control specimens embedded with steel, IHW-GFRP, and PHW-CFRP rebars, along with their corresponding CEB-FIP curves. The CEB-FIP curve of steel rebar

demonstrated notably accurate estimation of bond–slip behavior. However, for IHW-GFRP and PHW-CFRP rebars, the curves captured the ascending stage with acceptable accuracy, but they exhibited lower accuracy in predicting the descending and residual stages. The low accuracy for FRP rebars can be attributed to the assumption in the analytical model that after reaching peak bond stress, bond stress consistently decreases in a descending stage until reaching its minimum value. However, experimental curves of these rebars indicated that after reaching peak bond stress, bond stress sharply decreased in a descending stage before gradually decreasing in a residual stage. Consequently, the conventional CEB-FIP model for FRP rebars requires modification.

4.2. Modified CEB-FIP model

To address the observed discrepancies between the bond–slip curves of both types of FRP rebars and their corresponding CEB-FIP models, this study proposed a modification to the CEB-FIP model tailored specifically for FRP rebars. The disparities arise from the assumption made by the conventional CEB-FIP model that a single descending stage following the ascending stage is sufficient to describe bond–slip behavior. To resolve this limitation, the model was modified by dividing the descending stage of the conventional approach into two distinct stages: a descending stage and a residual stage. As illustrated in Fig. 15, the ascending stage of the CEB-FIP model aligns well with the experimental curves. Therefore, the ascending stage in the modified model is kept identical to that of the conventional CEB-FIP model, with the bond–slip relationship defined using Eq. (3a) and labeled as Eq. (4a) for the modified model. The descending stage of the modified model is defined by a linear relationship between bond stress and slippage, as mathematically expressed in

**Fig. 16.** Modified CEB-FIP model for FRP rebars.**Fig. 15.** Experimental bond–slip curves alongside CEB-FIP bond–slip curves.

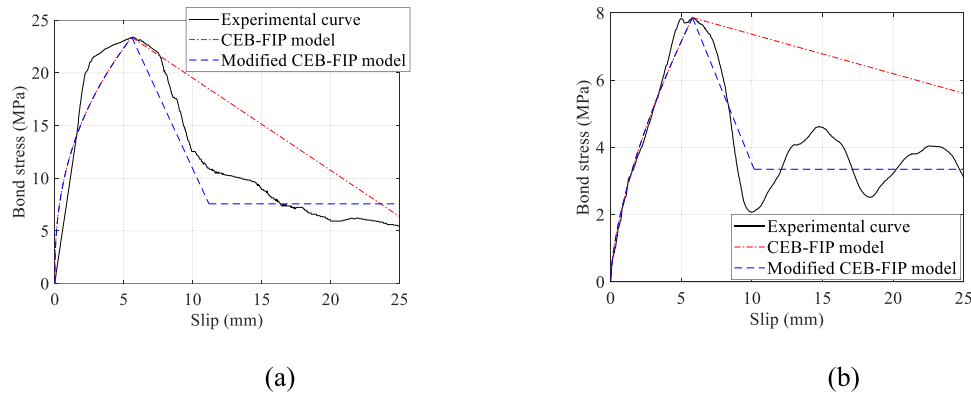


Fig. 17. Experimental bond-slip curves alongside conventional CEB-FIP and modified CEB-FIP bond-slip curves of control specimens embedded with (a) IHW-GFRP rebars and (b) PHW-CFRP rebars.

Eq. (4b). As shown in Fig. 16, this stage begins at the peak bond stress and its corresponding slippage, marking the end of the ascending stage, and ends at the turning point slippage, referred to as the residual bond stress corresponding slippage (s_f). Beyond this point, the reduction in bond stress becomes less pronounced as slippage increases. Similar to the determination of s_1 , s_2 , s_3 , and s_{max} in the CEB-FIP model, the value of s_f in the modified model was obtained through regression analysis of the experimental data. Following the descending stage, the residual stage of the modified model begins at the turning point slippage and extends to the maximum recorded slippage. For IHW-GFRP rebars, the bond stress variation during this stage is insignificant compared to the peak bond stress. In contrast, for PHW-CFRP rebars, although the bond stress variation is more pronounced, the second local peak has an amplitude less than half of the main peak bond stress. Consequently, in this modified model, the residual bond stress is assumed to be constant and is represented by Eq. (4c). This constant value is determined by averaging the bond stress values at multiple points within this stage for both types of rebars.

$$\text{ascending stage : } \tau = \tau_{max} \left(\frac{s}{s_{max}} \right)^\alpha \text{ for } 0 \leq s \leq s_{max} \quad (4a)$$

$$\text{descending stage : } \tau = \tau_{max} - (\tau_{max} - \tau_f) \left(\frac{s - s_{max}}{s_f - s_{max}} \right) \text{ for } s_{max} < s \leq s_f \quad (4b)$$

$$\text{residual stage : } \tau = \tau_f \text{ for } s_f < s \quad (4c)$$

The modified and conventional CEB-FIP bond-slip curves of control specimens embedded with FRP rebars are presented in Fig. 17. For control specimens embedded with IHW-GFRP rebars, the coefficients of

Table 7

Residual bond stresses and corresponding slippages obtained from pull-out test results.

No. of combined cycles	Rebar type	τ_f (MPa)	s_f (mm)
Control specimens	IHW-GFRP	7.57	11.18
	PHW-CFRP	3.35	10.16
1 combined cycle	IHW-GFRP	6.42	11.11
	PHW-CFRP	2.80	10.65
2 combined cycles	IHW-GFRP	5.78	10.25
	PHW-CFRP	2.00	10.42
3 combined cycles	IHW-GFRP	5.84	10.06
	PHW-CFRP	1.42	10.13
4 combined cycles	IHW-GFRP	5.34	9.43
	PHW-CFRP	1.29	10.33
5 combined cycles	IHW-GFRP	—	—
	PHW-CFRP	1.13	10.77

determination (R^2) for the CEB-FIP and modified CEB-FIP bond-slip curves were 0.521 and 0.885, respectively, while the normalized root mean squared deviations (NRMSEs) were 0.194 and 0.095, respectively. Similarly, for control specimens embedded with PHW-CFRP rebars, the coefficients of determination for the CEB-FIP and modified CEB-FIP bond-slip curves were -1.486 and 0.837, respectively, with corresponding NRMSEs of 0.339 and 0.087, respectively. The negative R^2 value for the CEB-FIP bond-slip curve of PHW-CFRP rebars indicates a significantly poor fit of the model to the experimental data. Although both models utilized experimentally derived parameters such as s_{max} and s_f , the bond-slip curves generated by the modified model showed better agreement with the experimental results for both types of FRP rebars.

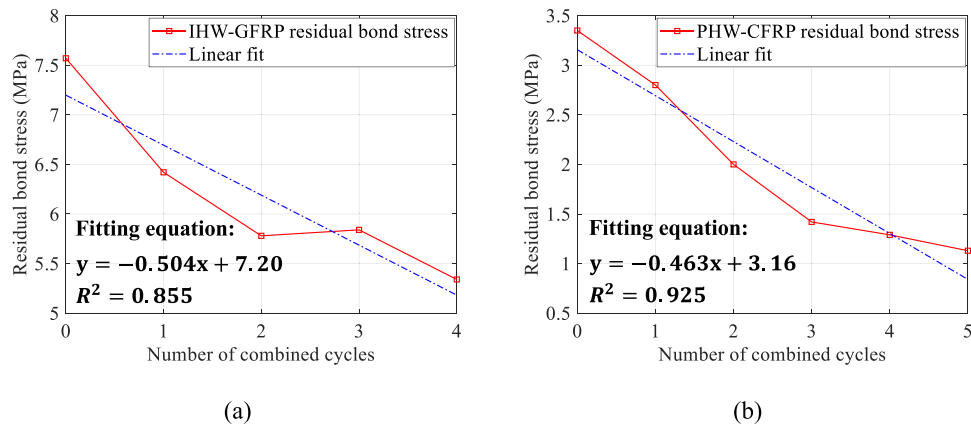


Fig. 18. Effect of combined FT and CI cycles on residual bond stress between concrete and (a) IHW-GFRP rebars and (b) PHW-CFRP rebars.

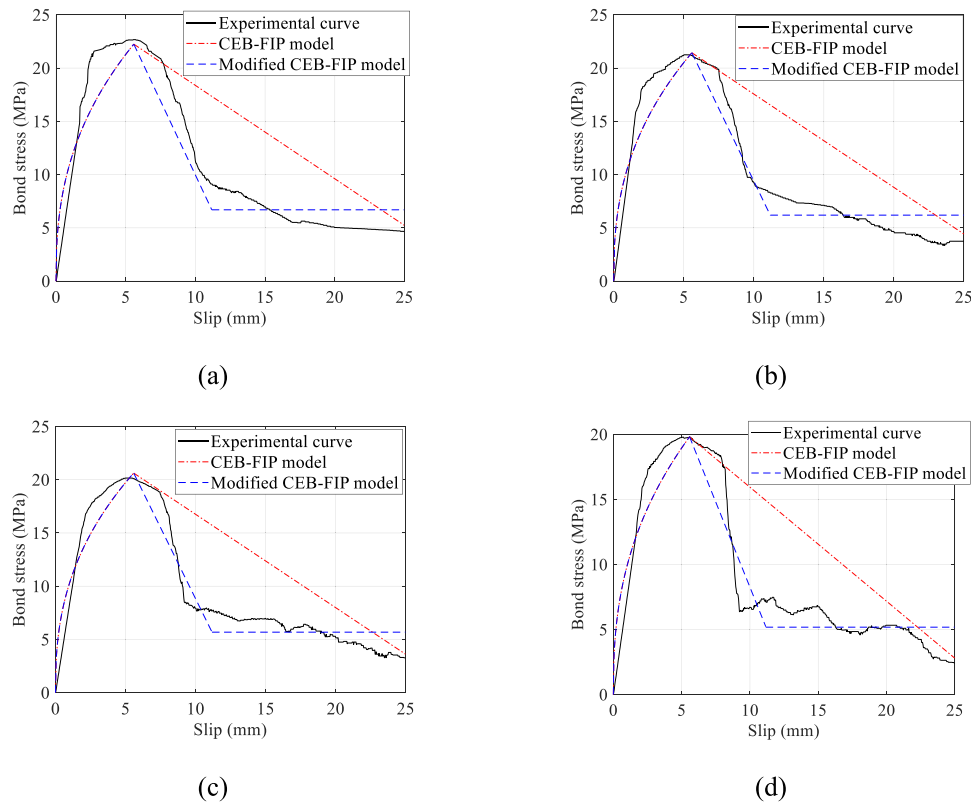


Fig. 19. Analytical prediction of the bond-slip relationship using conventional and modified CEB-FIP models for IHW-GFRP rebars embedded in concrete specimens subjected to (a) 1 combined cycle, (b) 2 combined cycles, (c) 3 combined cycles, and (d) 4 combined cycles.

Fig. 18 illustrates the relationship between the measured residual bond stress values and the combined FT and CI cycles. The graph shows a consistent reduction in residual bond stress for both IHW-GFRP and PHW-CFRP rebars as the number of cycles increases. The detailed values of residual bond stress and their corresponding slippages for both types of FRP rebars are provided in Table 7.

4.3. Bond-slip estimation using modified model

To analytically estimate the bond-slip behavior under the influence of combined FT and CI cycles, linearly fitted peak and residual bond stresses were applied to both conventional and modified CEB-FIP models. Given that the variations in slippages corresponding to peak and residual bond stresses, as well as the curvature parameters α , were minimal and showed no specific trend, the slippages and curvature parameters from the control specimens were used for this estimation. Figs. 19 and 20 depict experimental bond-slip curves, conventional CEB-FIP curves, and curves from the modified CEB-FIP model for IHW-GFRP and PHW-CFRP rebars subjected to combined FT and CI cycles. A quantitative evaluation of the models' accuracy was performed by calculating the coefficients of determination and NRMSDs, as summarized in Table 8. According to the results, the conventional CEB-FIP model showed poor performance in predicting the bond-slip relationships for both types of FRP rebars, with the highest coefficient of determination recorded at 0.49 and the lowest NRMSD at 0.216. In contrast, the modified CEB-FIP model demonstrated significantly better predictive capability, with the lowest coefficient of determination at 0.782 and the highest NRMSD at 0.136.

5. Conclusion

This research investigated the coupled effect of FT and CI cycles on bond-slip behavior of concrete specimens embedded with steel, IHW-

GFRP, and PHW-CFRP rebars. Additionally, the performance of the CEB-FIP analytical model in estimating bond-slip behavior was assessed. To enhance the accuracy of this model for FRP rebars, a modification was proposed and validated. Based on the results of this study, the following conclusions were drawn.

1. Due to the superior mechanical interlocking and frictional resistance from the rigid ribs and ridges on steel and IHW-GFRP rebars, control specimens embedded with these rebars showed much higher bond strengths. Conversely, control specimens embedded with PHW-CFRP rebars had much lower bond strength because their semi-deformed surfaces lacked sufficient rigidity.
2. The results of compressive strength tests demonstrated a consistent reduction in compressive strength of concrete specimens as the number of combined FT and CI cycles increased. This reduction was primarily attributed to propagation of cracks and deterioration of the concrete matrix.
3. Empirical observations on bond strength loss and failure modes indicated that the combined effect of FT and CI cycles primarily degraded the concrete matrix, with minimal impact on the surface condition of steel and IHW-GFRP rebars. However, for specimens with PHW-CFRP rebars, these cycles not only caused degradation of the concrete matrix but also notably affected the surface condition of the rebars by dislodging the adhered fiber bundles.
4. The comparison of experimental curves with those from the conventional CEB-FIP model showed a strong correlation for steel rebars but a weaker one for IHW-GFRP and PHW-CFRP rebars, due to the FRP model's assumption of a single descending stage after the ascending phase. To improve accuracy, this study proposed modifying the model to separate the residual stage from the descending stage for FRP rebars. The modified model achieved a good correlation with the experimental curves.

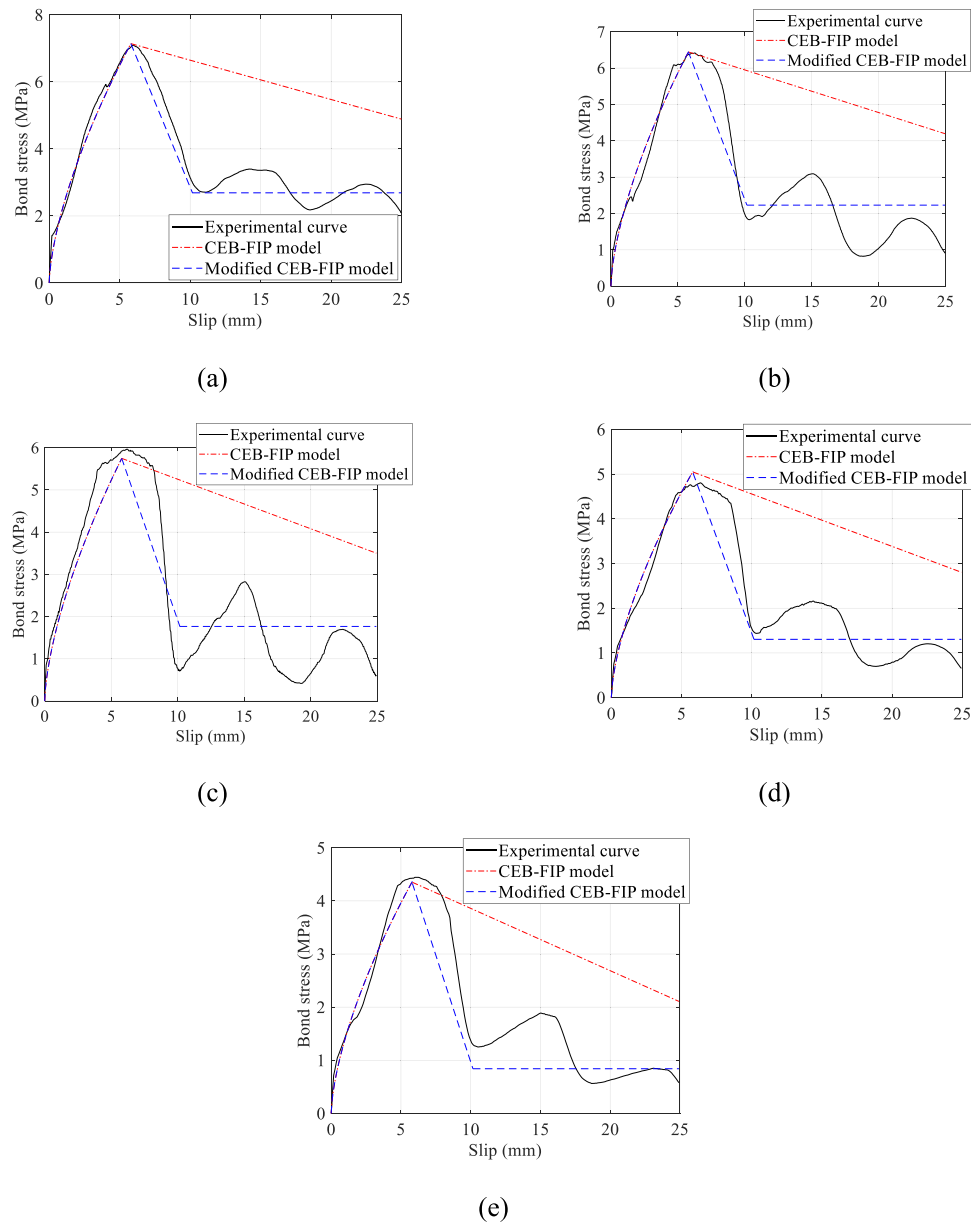


Fig. 20. Analytical prediction of the bond-slip relationship using conventional and modified CEB-FIP models for PHW-CFRP rebars embedded in concrete specimens subjected to (a) 1 combined cycle, (b) 2 combined cycles, (c) 3 combined cycles, (d) 4 combined cycles, and (e) 5 combined cycles.

Table 8

Coefficients of determination and NRMSDs for conventional and modified CEB-FIP models.

No. of combined cycles	Rebar type	CEB-FIP model		Modified CEB-FIP model	
		R^2	NRMSD	R^2	NRMSD
1 combined cycle	IHW-GFRP	0.490	0.216	0.894	0.098
	PHW-CFRP	-1.493	0.340	0.932	0.056
2 combined cycles	IHW-GFRP	0.440	0.223	0.915	0.087
	PHW-CFRP	-1.225	0.407	0.823	0.115
3 combined cycles	IHW-GFRP	0.425	0.221	0.925	0.080
	PHW-CFRP	-0.868	0.419	0.802	0.136
4 combined cycles	IHW-GFRP	0.438	0.224	0.904	0.093
	PHW-CFRP	-0.883	0.388	0.826	0.118
5 combined cycles	IHW-GFRP	—	—	—	—
	PHW-CFRP	-0.405	0.343	0.782	0.135

To further enhance the robustness and generalizability of the proposed model, future studies will focus on validating its performance across a variety of specimens reinforced with different types and sizes of

rebars, as well as various concrete mixture designs. This will ensure the model's applicability to a broader range of practical scenarios.

CRediT authorship contribution statement

Ali Azad: Writing – original draft, Visualization, Validation, Supervision, Investigation, Formal analysis. **Timur Zhylykaidar:** Writing – review & editing, Validation, Formal analysis, Data curation. **Jong-Han Lee:** Writing – review & editing, Validation, Supervision, Methodology, Funding acquisition, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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